Rudhira Metal Industries

ISO 9001-2015 CERTIFIED COMPANY

OFFICIAL TECHNICAL SPECIFICATION CATALOGUE

# BUTTWELD FITTINGS DIMENSIONS CATALOGUE

## DIMENSIONAL DATA & ENGINEERING STANDARDS

Seamless & Welded Pipe Fittings Dimensions to ANSI B16.9 - 2018

### COVERED TECHNICAL STANDARDS & SPECIFICATIONS

- ANSI B16.9 (2018) — Dimensions of Butt-Welding Fittings (1/2" NB to 36" NB)
- 90° Long Radius Elbows (A = 1.5D) & 90° Short Radius Elbows (N = 1.0D)
- 45° Elbows — Center to Face Dimensions (B)
- 180° Long Radius Returns (F, R) & 180° Short Radius Returns (G, M)
- Equal Tees (C, D) — Center to End Dimensions
- Buttweld Caps (E) & MSS SP-43 / B16.9 Stub-Ends (Short & Long Length L)
- Stainless Steel, Carbon Steel, Alloy Steel & Nickel Alloy Metallurgy Grades

RUDHIRA METAL INDUSTRIES — QUALITY ASSURANCE

ISO 9001:2015 Certified Manufacturing & Stockholding Facility

Material Test Certificates (EN 10204 3.1) • Positive Material Identification (PMI)

Third Party Inspection: Bureau Veritas, DNV, TUV, Lloyd's Register, PDIL, SGS

**RUDHIRA METAL INDUSTRIES**

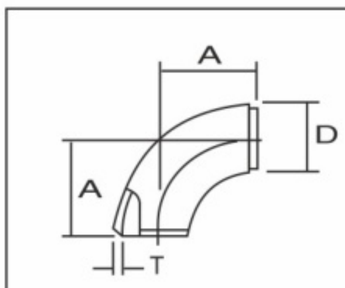
Commercial Office: Khetwadi, Mumbai - 400 004, Maharashtra, India

Branches: GIDC Ankleshwar (Gujarat) | Ranigunj, Hyderabad (Telangana)

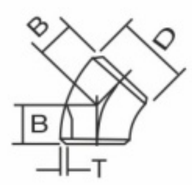
Phone: +91 98920 71909 / +91 99879 73244 • Email: sales@rudhirametal.com • Web: www.rudhirametal.com

Commercial Desk: +91 98920 71909 | +91 99879 73244 • sales@rudhirametal.com
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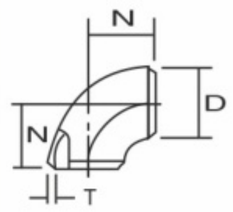

## DIMENSIONS OF BUTT-WELDING FITTING ANSI B 16.9 - 2018



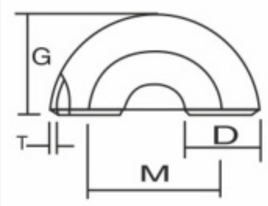
90°Elbow Long Radius



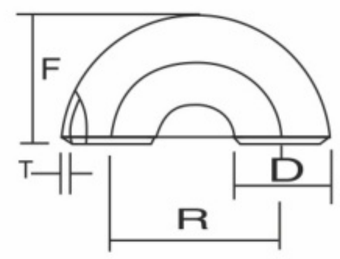
45°Elbow



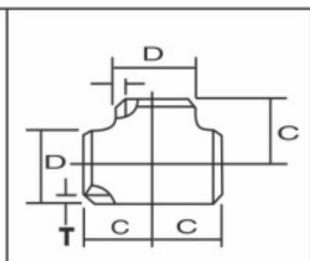
90°Elbow Short Radius



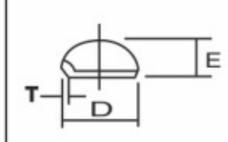
180°Return Short Radius



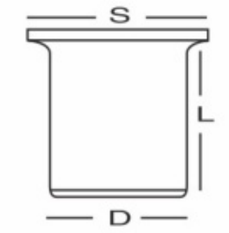
180°Return Long Radius



Equal Tee

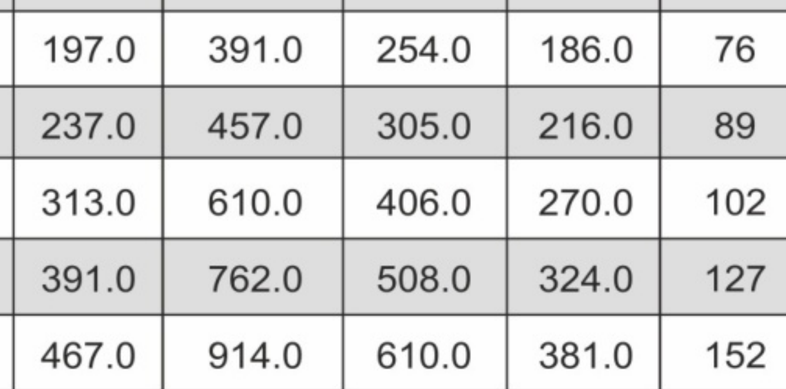


Caps



Stub-End

| Nominal Pipe Size | | Outside Diameter | Center to Face | | | | Back to Face | | | Center to Center | | | Length 'L' MSS SP 43 B16.9 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| INCH | NB | D | A R=1.5D | B | C | N R=1D | E | F | G | R | M | S | Short L | Long L |
| 1/2 | 15 | 21.3 | 38.00 | 16.0 | 25.0 | - | 25.0 | 48.0 | - | 76.0 | - | 35.0 | 51 | 76.0 |
| 3/4 | 20 | 26.7 | 38.00 | 19.0 | 29.0 | - | 25.0 | 51.0 | - | 76.0 | - | 49.0 | 51 | 76.0 |
| 1 | 25 | 33.4 | 38.00 | 22.0 | 38.0 | 25.0 | 38.0 | 56.0 | 41.0 | 76.0 | 51.0 | 51.0 | 51 | 102.0 |
| 1 1/4 | 32 | 42.2 | 48.00 | 25.0 | 48.0 | 32.0 | 38.0 | 70.0 | 52.0 | 95.0 | 64.0 | 64.0 | 51 | 102.0 |
| 1 1/2 | 40 | 48.3 | 57.00 | 29.0 | 57.0 | 38.0 | 38.0 | 83.0 | 62.0 | 114.0 | 76.0 | 73.0 | 51 | 102.0 |
| 2 | 50 | 60.3 | 76.00 | 35.0 | 64.0 | 51.0 | 38.0 | 106.0 | 81.0 | 152.0 | 102.0 | 92.0 | 64 | 152.0 |
| 2 1/2 | 65 | 73.0 | 95.00 | 44.0 | 76.0 | 64.0 | 38.0 | 132.0 | 100.0 | 190.0 | 127.0 | 105.0 | 64 | 152.0 |
| 3 | 80 | 88.9 | 114.00 | 51.0 | 86.0 | 76.0 | 51.0 | 159.0 | 121.0 | 229.0 | 152.0 | 127.0 | 64 | 152.0 |
| 3 1/2 | 90 | 101.6 | 133.00 | 57.0 | 95.0 | 89.0 | 64.0 | 184.0 | 140.0 | 267.0 | 178.0 | 140.0 | 76 | 152.0 |
| 4 | 100 | 114.3 | 152.0 | 64.0 | 105.0 | 102.0 | 64.0 | 210.0 | 159.0 | 305.0 | 203.0 | 157.0 | 76 | 152.0 |
| 5 | 125 | 141.3 | 190.0 | 79.0 | 124.0 | 127.0 | 76.0 | 262.0 | 197.0 | 391.0 | 254.0 | 186.0 | 76 | 203.0 |
| 6 | 150 | 168.3 | 229.0 | 95.0 | 143.0 | 152.0 | 89.0 | 313.0 | 237.0 | 457.0 | 305.0 | 216.0 | 89 | 203.0 |
| 8 | 200 | 219.1 | 305.0 | 127.0 | 178.0 | 203.0 | 102.0 | 414.0 | 313.0 | 610.0 | 406.0 | 270.0 | 102 | 203.0 |
| 10 | 250 | 273.1 | 381.0 | 159.0 | 216.0 | 254.0 | 127.0 | 518.0 | 391.0 | 762.0 | 508.0 | 324.0 | 127 | 254.0 |
| 12 | 300 | 323.9 | 457.0 | 190.0 | 254.0 | 305.0 | 152.0 | 619.0 | 467.0 | 914.0 | 610.0 | 381.0 | 152 | 254.0 |
| 14 | 350 | 355.6 | 533.0 | 222.0 | 279.0 | 356.0 | 165.0 | 711.0 | 533.0 | 1067.0 | 711.0 | 413.0 | 152 | 305.0 |
| 16 | 400 | 406.4 | 610.0 | 254.0 | 305.0 | 406.0 | 178.0 | 813.0 | 610.0 | 1219.0 | 813.0 | 470.0 | 152 | 305.0 |
| 18 | 450 | 457.0 | 686.0 | 286.0 | 343.0 | 457.0 | 203.0 | 914.0 | 686.0 | 1372.0 | 914.0 | 533.0 | 152 | 305.0 |
| 20 | 500 | 508.0 | 762.0 | 318.0 | 381.0 | 508.0 | 229.0 | 1016.0 | 762.0 | 1524.0 | 1016.0 | 584.0 | 152 | 305.0 |
| 22 | 550 | 559.0 | 838.0 | 343.0 | 419.0 | 559.0 | 254.0 | 1118.0 | 838.0 | 1676.0 | 1118.0 | 641.0 | 152 | 305.0 |
| 24 | 600 | 610.0 | 914.0 | 381.0 | 432.0 | 610.0 | 267.0 | 1219.0 | 914.0 | 1829.0 | 1219.0 | 692.0 | 152 | 305.0 |
| 26 | 650 | 660.0 | 991.0 | 405.0 | 495.0 | 660.0 | 267.0 | | | | | | | |
| 28 | 700 | 711.0 | 1067.0 | 438.0 | 521.0 | 771.0 | 267.0 | | | | | | | |
| 30 | 750 | 762.0 | 1143.0 | 470.0 | 559.0 | 762.0 | 267.0 | | | | | | | |
| 32 | 800 | 813.0 | 1219.0 | 502.0 | 597.0 | 813.0 | 267.0 | | | | | | | |
| 34 | 850 | 864.0 | 1295.0 | 533.0 | 635.0 | 864.0 | 267.0 | | | | | | | |
| 36 | 900 | 914.0 | 1372.0 | 565.0 | 673.0 | 914.0 | 267.0 | | | | | | | |



All Dimensions in Millimeters